

Computer Science & IT

Database Management System



Relational Model & Normal Forms

Lecture No. 07



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Recap of Previous Lecture



✓
Topic

Candidate key identification

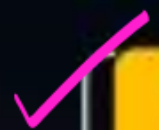


Topics to be Covered



Topic

Membership test



Topic

Relation between two FD sets



Topic

FD set of a sub-relation





Topic : Membership test

➤ Membership test is used to check whether a given FD is a member of given FD set or not.

➤ To check whether $X \rightarrow Y$ is a member of FD set F or not

(i.e., $F \models X \rightarrow Y$ or not)

$\equiv F$ implies $X \rightarrow Y$ or not
 $\equiv X \rightarrow Y$ is implied by F or not

$X \rightarrow Y$ can be inferred by FD set F or not

We first obtain X^+ (closure of X) w.r.t. FD set F .

If $Y \in X^+$, then $X \rightarrow Y$ is a member of FD set F
otherwise not a member of FD set F

#Q. Let FD set $F = \{ A \rightarrow B, B \rightarrow C \}$

Check whether $A \rightarrow C$ is a member of F or not?

$(A)^+$ w.r.t. $F = \{ A, B, C \}$

$C \in A^+$

$\therefore A \rightarrow C$ is implied by F .

#Q. Let FD set $F = \{ AB \rightarrow C, BC \rightarrow D \}$

Check whether $AB \rightarrow D$ is a member of F or not?

$$\overline{x} \quad \overline{y}$$

$(AB)^+ \text{ w.r.t. } F = \{A, B, C, D\}$

$$D \in (AB)^+$$

$\therefore AB \rightarrow D$ is a member of F

#Q. Let FD set $F = \{ AB \rightarrow C, C \rightarrow A \}$

Check whether $C \rightarrow B$ is a member of F or not?

$(C)^+ \text{ w.r.t. } F = \{ C, A, \}$

$B \notin (C)^+$

$\therefore C \rightarrow B$ is not a member of F .

$$F = \{fd_1, fd_2, \dots\}$$

$$G = \{fd'_1, fd'_2, \dots\}$$

$$F \subseteq G$$

$$G \subseteq F$$

$$F \subset G$$

$$G \subset F$$

$$F = G$$

$$F \not\subseteq G$$

$$G \not\subseteq F$$

$$F \not\subset G$$

$$G \not\subset F$$

if $A \subseteq B$ & $B \subseteq A$
then $A = B$

- Every element of set A is a member of set B
then $A \subseteq B$
- Every element of set B is a member of set A,
then $B \subseteq A$



Topic : Relationship between two FD sets

- Let F and G are any two FD sets.
- { If all the FDs of FD set F are member of FD set G, then $F \subseteq G$
 - (or) if G covers F, then $F \subseteq G$
 - (or) if all FDs of F are implied by G, then $F \subseteq G$
- If all the FDs of FD set G are member of FD set F, then $G \subseteq F$
 - (or) if F covers G, then $G \subseteq F$
 - (or) if all FDs of G are implied by F, then $G \subseteq F$
- If both $F \subseteq G$ and $G \subseteq F$ are true, then $F = G$

if $F \subseteq G$, but $G \not\subseteq F$, then $F \subset G$

if $G \subseteq F$, but $F \not\subseteq G$, then $G \subset F$

#Q. Consider the following FD sets

$F1 = \{A \rightarrow B, B \rightarrow C, AB \rightarrow D\}$ and $F2 = \{A \rightarrow B, B \rightarrow C, A \rightarrow C, A \rightarrow D\}$

Find the relationship between FD sets $F1$ and $F2$

Case ① Check if $F1$ Covers $F2$ } i.e. Check if all the FDs of $F2$ are member of $F1$ or not

FDs of $F2$	Closure w.r.t. $F1$	Remark	Conclusion
$A \rightarrow B$ $A \rightarrow C$ $A \rightarrow D$	$(A)^+ \text{ w.r.t } F1 = \{A, B, C, D\}$	$B \in A^+ \therefore$ Member of $F1$ $C \in A^+ \therefore$ Member of $F1$ $D \in A^+ \therefore$ Member of $F1$	All FDs of $F2$ are member of $F1$ $\therefore F2 \subseteq F1$ — eqn ①
$B \rightarrow C$	$(B)^+ \text{ w.r.t. } F1 = \{B, C\}$	$C \in B^+ \therefore$ Member of $F1$	

#Q. Consider the following FD sets

$F1 = \{A \rightarrow B, B \rightarrow C, AB \rightarrow D\}$ and $F2 = \{A \rightarrow B, B \rightarrow C, A \rightarrow C, A \rightarrow D\}$

Find the relationship between FD sets $F1$ and $F2$

Case ② Check if $F2$ covers $F1$ { i.e. Check if all the FDs of $F1$ are member of $F2$ or not }

FDs of $F1$	Closure w.r.t. $F2$	Remark	Conclusion
$A \rightarrow B$	$(A)^+ \text{ w.r.t. } F2 = \{A, B, C, D\}$	$B \in (A)^+ \therefore$ Member of $F2$	All FDs of $F1$ are member of $F2$ $\therefore F1 \subseteq F2$ — eq ②
$B \rightarrow C$	$(B)^+ \text{ w.r.t. } F2 = \{B, C\}$	$C \in (B)^+ \therefore$ Member of $F2$	
$AB \rightarrow D$	$(AB)^+ \text{ w.r.t. } F2 = \{A, B, C, D\}$	$D \in (AB)^+ \therefore$ Member of $F2$	

By eqⁿ ① & eqⁿ ②

$$F_1 = F_2$$

#Q. Consider the following FD sets

$F1 = \{A \rightarrow B, B \rightarrow C, A \rightarrow C\}$ and $F2 = \{A \rightarrow B, B \rightarrow C, A \rightarrow D\}$

Find the relationship between FD sets $F1$ and $F2$

FDs of $F1$

$A \rightarrow B$
 $B \rightarrow C$ } Directly present in $F2$

$A \rightarrow C$] $(A)^+$ wrt $F2 = \{A, B, C, D\}$
 $C \in (A)^+ \therefore$ Member of $F2$
 $F1 \subseteq F2$ — eqn ①

FDs of $F2$

$A \rightarrow B$
 $B \rightarrow C$ } Directly present in $F1$

$A \rightarrow D$
 $(A)^+$ wrt $F1 = \{A, B, C\}$
 $D \notin (A)^+ \therefore$ Not a member of $F1$

$\therefore F2 \not\subseteq F1$
eqn ②

$$F_1 \subseteq F_2 \text{ by } F_2 \not\subseteq F_1$$

$$\therefore F_1 \subset F_2$$

If the given options are

- (a) $F_1 \subseteq F_2$ ✓
- (b) $F_1 \neq F_2$ ✓
- (c) $F_1 \subset F_2$ ✓
- (d) $F_1 = F_2$

If MSQ question

$$\underline{\text{Ans}} = \underline{A, B, C}$$

If MCQ question.

$$\text{Ans} = \textcircled{C}$$

#Q. Consider the following FD sets

$F1 = \{A \rightarrow B, AB \rightarrow C, D \rightarrow AC, D \rightarrow E\}$ and

$F2 = \{A \rightarrow BC, D \rightarrow AE\}$

Find the relationship between FD sets $F1$ and $F2$

FDs of $F1$	Member of $F2$
$A \rightarrow B$	✓
$AB \rightarrow C$	✓
$D \rightarrow AC$	✓
$D \rightarrow E$	✓

∴ $F1 \subseteq F2$

FDs of $F2$	Member of $F1$
$A \rightarrow BC$	✓
$D \rightarrow AE$	✓

∴ $F2 \subseteq F1$

Ans =

$F1 = F2$



Topic : FD set of a sub-relation

- Let R be a relation with FD set F . We can use the concept of membership test to obtain the FD set of a sub-relation of relation R .

eg $R(ABC)$

$F = \left\{ \begin{array}{l} A \rightarrow B \\ B \rightarrow C \end{array} \right\}$



if $R_1(AC)$ is a subrelation of relation R , then identify FD set F_1 w.r.t. subrelation R_1

w.r.t. FD set F , $(A)^+ = \{A, B, C\}$ i.e. ~~$A \rightarrow ABC$~~

$(C)^+ = \{C\}$ i.e. ~~$C \rightarrow C$~~

$\therefore$ FDs of FD set $F_1 = \{A \rightarrow C\}$

#Q. Consider a relational schema $R(A, B, C, D, E)$ with FD set

$$F = \{A \rightarrow B, AB \rightarrow C, D \rightarrow AC, D \rightarrow E\}$$

Find the FD set F_1 for sub-relation $R_1(A, B, E)$ of $R(A, B, C, D, E)$.

Also find candidate keys for the sub-relation $R_1(A, B, E)$.

* R_1 is a sub-relation with three attributes A, B & E , these attributes are projected from relation R , so the relationship that existed b/w these three attributes in relation R will hold true in relation $R_1(A, B, E)$ as well

$$(A)^+ \text{ w.r.t. } F = \{A, B, C\} \Rightarrow A \rightarrow A B C$$

$$(B)^+ \text{ w.r.t. } F = \{B\} \Rightarrow B \rightarrow B$$

$$(E)^+ \text{ w.r.t. } F = \{E\} \Rightarrow E \rightarrow E$$

trivial information

Not in R_1

$$(AB)^+ \text{ w.r.t. } F = \{A, B, C\} \Rightarrow AB \rightarrow ABC$$

$$(AE)^+ \text{ w.r.t. } F = \{A, B, C, E\} \Rightarrow AE \rightarrow ABCE$$

$$(BE)^+ \text{ w.r.t. } F = \{B, E\} \Rightarrow BE \rightarrow BE$$

trivial

Not in R_1

Not in R_1

$$(ABE)^+ = \{A, B, C, E\}$$

$$ABE \rightarrow ABCE$$

Not in R_1

Non trivial FDs which will hold true in relation $R_1(A, B, E)$ are,

$$A \rightarrow B$$

$$AE \rightarrow B$$

$\therefore F_1 = \left\{ \begin{array}{l} A \rightarrow B \\ AE \rightarrow B \end{array} \right\}$ is the FD set w.r.t. $R_1(A, B, E)$

$$(AE)^+ = \{A, E, B\}$$

all attributes of $R_1 \therefore$ S.K.

$$(A)^+ = \{A, B\}$$

$$(E)^+ = \{E\}$$

No proper subset is a Super Key

$\therefore \boxed{AE}$ is a Candidate Key

Prime Attributes = $\{A, E\}$
 $\therefore \boxed{AE}$ is the Only Candidate Key.

#Q. Consider a relational schema $R(A,B,C,D,E,F)$ with FD set

$$F = \{AB \rightarrow C, B \rightarrow D, BC \rightarrow A, D \rightarrow EF\}$$

Find the FD set F_1 for sub-relation $R_1(A,B,C,D)$ of $R(A,B,C,D,E,F)$.

Also find candidate keys for the sub-relation $R_1(A,B,C,D)$.

$$(A)^+ \text{ wrt } F = \{A\} \quad A \rightarrow A$$

$$(B)^+ \text{ wrt } F = \{B, D, E, F\} \quad B \rightarrow D$$

$$(C)^+ \text{ wrt } F = \{C\} \quad C \rightarrow C$$

$$(D)^+ \text{ wrt } F = \{D, E, F\} \quad D \rightarrow EF$$

$$(AB)^+ = \{A, B, C, D, E, F\} \therefore AB \rightarrow CD$$

$$(AC)^+ = \{A, C\}$$

$$(AD)^+ = \{A, D, E, F\}$$

$$(BC)^+ = \{B, C, D, A, E, F\} \therefore BC \rightarrow AD$$

$$(BD)^+ = \{B, D, E, F\}$$

$$(CD)^+ = \{C, D, E, F\}$$

$$(ABC)^+ = \{\cancel{A}, \cancel{B}, \cancel{C}, D, E, F\}$$

$$\therefore \boxed{ABC \rightarrow D}$$

$$(ABD)^+ = \{\cancel{A}, \cancel{B}, C, \cancel{D}, E, F\}$$

$$\therefore \boxed{ABD \rightarrow C}$$

$$(ACD)^+ = \{\cancel{A}, \cancel{C}, \cancel{D}, E, F\}$$

$$(BCD)^+ = \{\cancel{B}, \cancel{C}, \cancel{D}, A, E, F\}$$

$$\therefore \boxed{BCD \rightarrow A}$$

FD set F_1 w.r.t. sub-relation $R_1(A, B, C, D)$ is

$F_1 = \left\{ \begin{array}{l} B \rightarrow D \\ AB \rightarrow CD \\ BC \rightarrow AD \\ ABC \rightarrow D \\ ABD \rightarrow C \\ BCD \rightarrow A \end{array} \right\}$

$\boxed{AB}$ & $\boxed{BC}$ are
the only two candidate
keys of R_1

$(AB)^+ = \{A, B, C, D\}$
all attributes $\therefore AB$ is a S.K.
 $(A)^+ = \{A\}$
 $(B)^+ = \{B, D\}$
 $\downarrow$
and $\boxed{AB}$ is a C.K.
Prime Attributes: $\{A, B\}$

$BC \rightarrow A$
 $\therefore$ Replace A by BC
 BC is a Super key $\Rightarrow$
and $\boxed{BC}$ is a Candidate Key
Prime Attributes: $\{A, B, C\}$
 $(B)^+ = \{B, D\}$
 $(C)^+ = \{C\}$

⑤ Tomorrow's Topic :-

Homework



Read about

Minimal Cover (Canonical Cover)



Minimal Cover of FD set

(or)

Canonical Cover of FD set

(or)

Irreducible set of FDs w.r.t. given FD set.

THANK - YOU